

SL-Assignment-2

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1. Implement basic sorting algorithms (bubble sort, selection sort) on arrays.

Program:-

Bubble Sort-

```
package slassignment2;

import java.util.Scanner;

public class BubbleSort {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter number of elements: ");
        int n=sc.nextInt();
        int []arr=new int[n];

        System.out.println("Enter elements:");
        for (int i=0; i<n; i++) {
            arr[i]=sc.nextInt();
        }

        for (int i=0; i<n-1; i++) {
            for (int j=0; j<n-i-1; j++) {
                if (arr[j]>arr[j+1]) {
                    int temp=arr[j];
                    arr[j]=arr[j+1];
                    arr[j+1]=temp;
                }
            }
        }

        System.out.println("Sorted array:");
        for (int num : arr) {
            System.out.print(num+" ");
        }

        sc.close();
    }
}
```

Selection Sort-

```
package slassignment2;

import java.util.Scanner;

public class SelectionSort {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter number of elements: ");
        int n=sc.nextInt();
        int []arr=new int[n];

        System.out.println("Enter elements:");
        for (int i=0; i<n; i++) {
            arr[i]=sc.nextInt();
        }

        for(int i=0;i<n-1;i++){
            for(int j=i+1;j<n;j++){
                if(arr[i]>arr[j]){
                    int temp=arr[j];
                    arr[j]=arr[i];
                    arr[i]=temp;
                }
            }
        }

        System.out.println("Sorted array:");
        for (int num : arr) {
            System.out.print(num+" ");
        }

        sc.close();
    }
}
```

2. Leap year check

Check if a given year is a leap year.

Program:-

```
package slassignment2;

import java.util.Scanner;

public class LeapYear{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter the year: ");
        int year=sc.nextInt();

        if((year%4==0 && year%100!=0) || year%400==0){
            System.out.println("Its Leap Year");
        }
        else{
            System.out.println("Not a Leap Year");
        }
        sc.close();
    }
}
```

3. Palindrome Check

Check if a given String or a number series is a Palindrome.

Program:-

```
package slassignment2;

import java.util.Scanner;

public class Palindrome{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter the number and string: ");
        int num=sc.nextInt();
        String str=sc.next();
        //for number
        int result=0,digit;
        int temp=num;
        while(temp!=0){
```

```
        digit=temp%10;
        result=result*10+digit;
        temp=temp/10;
    }
    if(num==result){
        System.out.println(num+" is palindrome");
    }
    else{
        System.out.println(num+" is not a palindrome");
    }

    //for string
    StringBuilder newStr=new StringBuilder();
    for(int i=str.length()-1;i>=0;i--){
        char ch=str.charAt(i);
        newStr.append(ch);
    }
    if(str.equals(newStr.toString())){
        System.out.println(str+" is palindrome");
    }
    else{
        System.out.println(str+" is not a palindrome");
    }
}
}
```